

NONCENTRAL DISTRIBUTIONS (LINEAR CASE) OF SOME TEST CRITERIA IN MULTIVARIATE ANALYSIS

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1. Introduction and Summary

Let $A(p \times p)$ be distributed according to a Wishart distribution with m degrees of freedom and population covariance matrix I and $B(p \times p)$ be distributed independently according to a noncentral Wishart distribution with n degrees of freedom and same covariance matrix I , corresponding to the linear case. We shall adopt the notation $A(p \times p) \sim W(I, m)$ and $B(p \times p) \sim W(I, n, \lambda^2)$ where λ^2 is the noncentrality parameter. If $A = CLC'$, $A+B = CC'$ and C is a lower triangular matrix then the density of

$$(1.1) \quad L = C^{-1}AC^{-1'}$$

is called the noncentral Multivariate Beta type I distribution. Its density is given by

$$(1.2) \quad f(L) = C|L|^{\frac{1}{2}(m-p-1)}|I-L|^{\frac{1}{2}(n-p-1)}e^{-\frac{1}{2}\lambda^2} \\ \cdot {}_1F_1\left(\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\lambda^2(1-\ell_{11})\right), L > 0, I-L > 0$$

and where ℓ_{11} is the top lefthand element of L .

Further we have

$$(1.3) \quad C = \Gamma_p\left(\frac{1}{2}(m+n)\right)/(\Gamma_p\left(\frac{1}{2}n\right)\Gamma_p\left(\frac{1}{2}m\right))$$

$$(1.4) \quad \Gamma_p\left(\frac{1}{2}n\right) = \pi^{-\frac{1}{2}pn} \frac{\Gamma(p-1)\prod_{i=1}^p \Gamma\left(\frac{1}{2}(n-i+1)\right)}{1}$$

and where in general the hypergeometric function is defined by

$$(1.5) \quad {}_3F_t(a_1, a_2, \dots, a_s; b_1, b_2, \dots, b_t; \kappa^2) = \sum_{j=0}^{\infty} \frac{(a_1)_j (a_2)_j \dots (a_s)_j}{(b_1)_j (b_2)_j \dots (b_t)_j} \frac{(\kappa^2)^j}{j!}$$

where

$$(1.6) \quad (a)_j = a(a+1)\dots(a+j-1) = \Gamma(a+j)/\Gamma(a)$$

The multivariate Dirichlet distributions are derived in a similar way.

Let $A_j \sim W(I, m_j)$, $j=1, 2, \dots, k$ (independent) and let B be independently distributed $W(I, n, \lambda^2)$.

The joint density of $(L_1, L_2, \dots, L_k)$ where

$$(1.7) \quad L_j = (\sum_{j=1}^k A_j + B)^{-1/2} A_j (\sum_{j=1}^k A_j + B)^{-1/2}; \quad j=1, 2, \dots, k$$

is the noncentral multivariate Dirichlet type 1 distribution with density

$$(1.8) \quad D(m_1, m_2, \dots, m_k; n) \prod_{j=1}^k |L_j|^{1/2(m_j - p - 1)} \\ \cdot |I - \sum_{j=1}^k L_j|^{1/2(n - p - 1)} e^{-1/2\lambda^2} {}_1F_1(1/2(m+n); 1/2n; 1/2\lambda^2 \\ \cdot (I - \sum_{j=1}^k L_j)) \cdot L_j > 0; 1 - \sum_{j=1}^k L_j > 0$$

where

$$(1.9) \quad D(m_1, m_2, \dots, m_k; n) = \Gamma_p(1/2(m+n)) / (\Gamma_p(1/2n) \prod_{j=1}^k \Gamma_p(1/2m_j))$$

and

$$m = \sum_{j=1}^k m_j.$$

The multivariate Beta distributions and the multivariate Dirichlet distributions arise in various problems in multivariate analysis. Various test criteria for testing hypotheses in multivariate analysis depend on these distributions (Troskie (1971)), some of these being $|L|$, $|I-L|$ and $|I - \sum_{j=1}^k L_j|$.

The likelihood ratio test for testing the hypothesis that the means of several normal distributions with common covariance matrix are equal reduces to (Anderson (1958), Troskie (1966) and Gupta (1971))

$$(1.10) \quad \Lambda = |L|$$

By means of the method of convolutions, Schatzoff (1966) and Pillai and Gupta

(1969) derived the central distributions of the likelihood ratio statistic $|L|$. Gupta (1971) derived the noncentral distribution (linear case) of Wilks' likelihood ratio statistic in MANOVA and also explicit expressions for the density for $p = 2, 3, 4$ and 5. In this paper similar expressions are obtained for the test criteria $|I-L|$ and $|I - \sum_{j=1}^k L_j|$ in both the real and complex case.

Wilks' likelihood ratio criterion $|L|$ and the statistic $|I-L|$ can also be expressed in terms of certain multiple correlation coefficients and we also indicate the results for tests based on the Generalised multiple correlation matrix R (real case) and the Generalised multiple coherence matrix (complex case).

2. The distributions of $|I-L|$ and $|I - \sum_{j=1}^k L_j|$ in the noncentral linear case

The h -th moment of $|I-L|$ is given by (de Waal (1968))

$$\begin{aligned} (2.1) \quad E(|I-L|^h) &= (\Gamma_p(\frac{1}{2}(m+n))\Gamma_p(\frac{1}{2}n+h)/(\Gamma_p(\frac{1}{2}n)\Gamma_p(\frac{1}{2}(m+n)+h))) \\ &\quad \cdot e^{-\frac{1}{2}\lambda^2} {}_2F_2(\frac{1}{2}n+h, \frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}(m+n)+h, \frac{1}{2}\lambda^2) \\ &= E(\prod_{i=1}^p X_i)^h \end{aligned}$$

where X_1 has the density

$$\begin{aligned} (2.2) \quad \beta_1(x_1, \lambda^2, n, m, 1) &= (B(\frac{1}{2}n, \frac{1}{2}m))^{-1} x_1^{\frac{1}{2}n-1} (1-x_1)^{\frac{1}{2}m-1} \\ &\quad \cdot e^{-\frac{1}{2}\lambda^2} {}_1F_1(\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\lambda^2 x_1), \quad 0 < x_1 < 1 \end{aligned}$$

and $X_i (i=2, 3, \dots, p)$ has the density

$$(2.3) \quad \beta_1(x_i, n+1-i, m) = (B(\frac{1}{2}(n+1-i), \frac{1}{2}m))^{-1} x_i^{\frac{1}{2}(n+1-i)-1} (1-x_i)^{\frac{1}{2}m-1}, \quad 0 < x_i < 1$$

and where

$$(2.4) \quad B(\frac{1}{2}n, \frac{1}{2}m) = \Gamma(\frac{1}{2}n)\Gamma(\frac{1}{2}m)/\Gamma(\frac{1}{2}(m+n)).$$

The moments of $|I - \sum_{j=1}^k L_j|$ coincide (Troskie (1972)) with the moments of $|I-L|$ expect that in the former case $m = \sum_{j=1}^k m_j$. Writing $W_{p,m,n}$ for $|I-L|$ and $|I - \sum_{j=1}^k L_j|$, the value of m depending on the statistic under consideration, we have from (2.1), (2.2) and (2.3) the following theorem.

Theorem 2.1 (Gupta (1971))

In the noncentral linear case the statistic $W_{p,m,n}$ is distributed as $\Pi_{i=1}^p X_i$ where the $X_i (i=1,2,3,\dots,p)$ are independent, X_1 having the density $\beta_1(x_1; \lambda^2, n, m, 1)$ and the $X_i (i=1,2,3,\dots,p)$ the density $\beta_1(x_i; n+1-i, m)$.

Application of the duplication formula (Anderson (1958)) gives rise to the following theorem.

Theorem 2.2 (Gupta (1971))

In the noncentral linear case $W_{2r,m,n}$ is distributed like $X_1 X_{2r} \Pi_{i=1}^{r-1} Y_i^2$ where the $Y_i (i=1,2,\dots,r-1)$ are independent and Y_i has the density $\beta_1(y_i; 2(n-2i), 2m)$ and X_1 and X_{2r} are independently distributed with densities given by (2.2) and $\beta_1(x_{2r}; n+1-p, m)$ respectively.

$W_{2s+1,m,n}$ is distributed as $X_1 \Pi_{i=1}^s Y_i^2$ where the $Y_i (i=1,2,\dots,s)$ are independent and Y_i has the density $\beta_1(y_i; 2(n-2i), 2m)$ and X_1 is independently distributed with density given by (2.2).

The expressions for the density $W_{p,m,n}$ are derived as follows:

Having that

$$(2.5) \quad {}_1F_1(\tfrac{1}{2}(m+n); \tfrac{1}{2}n; \tfrac{1}{2}\lambda^2 x_1) = \sum_{j=0}^{\infty} \frac{(\tfrac{1}{2}(m+n))_j (\tfrac{1}{2}\lambda^2)^j}{(\tfrac{1}{2}n)_j j!} x_1^j$$

and substituting (2.5) in (2.2) gives

$$(2.6) \quad e^{-\frac{1}{2}\lambda^2} (B(\tfrac{1}{2}m, \tfrac{1}{2}n))^{-1} \sum_{j=0}^{\infty} a_j x_1^{\frac{1}{2}n+j-1} (1-x_1)^{\frac{1}{2}m-1}$$

where

$$(2.7) \quad a_j = (\tfrac{1}{2}(m+n))_j (\tfrac{1}{2}\lambda^2)^j / ((\tfrac{1}{2}n)_j j!)$$

For m even, substituting the binomial expansion of $(1-x_1)^{\frac{1}{2}m-1}$ in (2.6) and then making the transformation $y_1 = -\log x_1$ gives

$$(2.8) \quad e^{-\frac{1}{2}\lambda^2} (B(\tfrac{1}{2}m, \tfrac{1}{2}n))^{-1} \sum_{j=0}^{\infty} a_j \sum_{k=0}^b (-1)^k \binom{b}{k} \exp(-y_1(\tfrac{1}{2}n+j+k)), \quad y_1 \geq 0.$$

For m odd the binomial expansion is an infinite series and the distribution problems for this case are not discussed in this paper.

Similarly applying the binomial expansion to $(1-x_i)^{\frac{1}{2}m-1}$ and making the transformation $y_i = -\log x_i$ in (2.3) gives

$$(2.9) \quad (B(\tfrac{1}{2}m, \tfrac{1}{2}(n+1-i)))^{-1} \sum_{\ell=0}^b (-1)^\ell \binom{b}{\ell} \exp(-\tfrac{1}{2}y_1(n+1+2\ell-i)), \quad y_i \geq 0$$

Further from Theorem 2.2 the density of the random variable $Z_i = X_{2i} X_{2i+1}$ can be derived (Gupta (1971)) and then on making the transformation $Y_i' = -\log Z_i$ we obtain the following density.

$$(2.10) \quad f_{Y_i'}(y_i') = (2B(n-i, m))^{-1} \sum_{\ell=0}^{m-1} (-1)^\ell \binom{m-1}{\ell} \exp(-\frac{1}{2} y_i' (n+\ell-2i)), \quad y_i' \geq 0$$

From theorem 2.2 it follows that for $p = 2r$

$$(2.11) \quad -\log W = Y_1 + \sum_{i=1}^{r-1} y_i' + Y_{2r}$$

and for $p = 2s+1$

$$(2.12) \quad -\log W = Y_1 + \sum_{i=1}^s Y_i'$$

From (2.10) and (2.11), $-\log W$ is distributed as the sum of independently distributed random variables and therefore it is clear that expressions for the density of $W_{p,m,n}$ can be obtained by taking successive convolutions, provided, as Gupta (1971) points out, the procedure yields expressions which can be integrated easily at each stage. The expressions for the densities of $|I-L|$ and $|I-\Sigma L_j|$ are identical except that in the latter $m = \sum_{j=1}^k m_j$. Expressions for the densities and the cdf's of these two statistics for the case $p = 2, 3, 4$ and 5 are listed below.

Detailed derivations of these expressions can be found in (Money (1972)).

Case 1

For $p = 2$ the density of W is given by

$$(2.13) \quad K e^{-\frac{1}{2} \lambda^2} w^{\frac{1}{2} n - 1} \sum_{j=0}^{\infty} a_j \left\{ \sum_{\ell, k=0}^b (-1)^{\ell+k} \binom{b}{k} \binom{b}{\ell} (w^{j+k} - w^{\ell-\frac{1}{2}}) / (2\ell-2j-2k-1) \right\},$$

$$0 \leq w \leq 1$$

and the cumulative distribution function is given by

$$(2.14) \quad 2K e^{-\frac{1}{2} \lambda^2} w^{\frac{1}{2} n} \sum_{j=0}^{\infty} a_j \left\{ \sum_{\ell, k=0}^b (-1)^{\ell+k} \binom{b}{k} \binom{b}{\ell} [w^{j+k} / (n+2j+2k) - w^{\frac{1}{2}(2\ell-1)} / (n+2\ell-1)] / (2\ell-2j-2k-1) \right\},$$

$$0 \leq w \leq 1$$

where

$$(2.15) \quad b = \frac{1}{2}m-1$$

$$(2.16) \quad K = 2(B(\frac{1}{2}m, \frac{1}{2}(n-1)) B(\frac{1}{2}m, \frac{1}{2}n))^{-1}$$

$$(2.17) \quad a_j = (\frac{1}{2}(m+n))_j (\frac{1}{2}\lambda^2)^j / ((\frac{1}{2}n)_j j!)$$

Case 2

For $p = 3$ the density of W is given by

$$(2.18) \quad K e^{-\frac{1}{2}\lambda^2} w^{\frac{1}{2}n-1} \sum_{j=0}^{\infty} a_j \{ \sum_{k=0}^b (-w)^k w^j \binom{b}{k} \binom{m-1}{2j+2k+2} (-\log w) \\ + 2 \sum_{k=0}^b \sum_{\ell \neq 2k+2j+2}^{m-1} ((-1)^{\ell+k} / (\ell-2k-2j-2)) \binom{m-1}{\ell} \binom{b}{k} \\ (w^{j+k} - w^{\frac{1}{2}\ell-1}) \}, \quad 0 \leq w \leq 1$$

and the cumulative distribution function is given by

$$(2.19) \quad 2K e^{-\frac{1}{2}\lambda^2} w^{n/2} \sum_{j=0}^{\infty} a_j \{ \sum_{k=0}^b ((-1)^k / (n+2k+2j)^2) \binom{b}{k} \binom{m-1}{2j+2k+2} w^{k+j} \\ \cdot (2 - (n+2k+2j) \log w) + 2 \sum_{k=0}^b \sum_{\ell \neq 2k+2j+2}^{m-1} ((-1)^{\ell+k} / \\ \cdot (\ell-2j-2k-2)) \binom{m-1}{\ell} \binom{b}{k} [w^{k+j} / (n+2k+2j) - w^{\frac{1}{2}(\ell-2)} / (n+\ell-2)] \}, \quad 0 \leq w \leq 1$$

where

$$(2.20) \quad K = (2B(\frac{1}{2}m, \frac{1}{2}n) B(\frac{1}{2}(n-2), \frac{1}{2}m))^{-1}$$

Case 3

For $p = 4$ the density of W is given by

$$(2.21) \quad K e^{-\frac{1}{2}\lambda^2} w^{\frac{1}{2}n-1} \sum_{j=0}^{\infty} a_j \{ T_{1j} + 2T_{2j} \}$$

where

$$(2.22) \quad T_{1j} = \sum_{k,t} ((-1)^{k+t} / (2t-2j-2k-3)) w^{j+k} f(o, t, k) \{ \binom{m-1}{2j+2k+2} \\ \cdot ((2w^{t-j-k-3/2} - 2) / (2t-2j-2k-3) - \log w) - \binom{m-1}{2t-1} w^{t-k-j-3/2} \log w \}$$

and

$$(2.23) \quad T_{2j} = \sum_k \sum_{\ell, \ell \neq 2j+2k+2} ((-1)^{\ell+k}/(\ell-2j-2k-2)) f(\ell, o, k) \\ \cdot \{ \sum_t ((-1)^t/(2t-2j-2k-3)) \binom{b}{t} (w^{j+k} - w^{t-3/2}) \\ - \sum_{t, t \neq \frac{1}{2}(\ell+1)} ((-1)^t/(2t-\ell-1)) \binom{b}{t} (w^{\frac{1}{2}\ell-1} - w^{t-3/2}) \}$$

and the cumulative distribution function is given by

$$(2.24) \quad 2Ke^{-\frac{1}{2}\lambda^2} w^{\frac{1}{2}n} \sum_{j=0}^{\infty} a_j (T_{1j}^* + 2T_{2j}^*)$$

where

$$(2.25) \quad T_{1j}^* = \sum_{k,t} ((-1)^{k+t}/(2t-2j-2k-3)) f(o, t, k) w^{j+k} \\ \cdot \left\{ \binom{m-1}{2j+2k+2} [((2w^{t-j-k-3/2}/(n+2t-3) - 2/(n+2j+2k))/(2t-2j-2k-3) \right. \\ \left. + (2-(n+2j+2k)\log w)/(n+2j+2k)^2) \right] \\ \left. + \binom{m-1}{2t-1} \{ (2-(n+2t-3)\log w) w^{t-j-k-3/2} / (n+2t-3)^2 \} \right\}$$

and

$$(2.26) \quad T_{2j}^* = \sum_k \sum_{\ell, \ell \neq 2j+2k+2} ((-1)^{\ell+k}/(\ell-2j-2k-2)) f(\ell, o, k) \\ \cdot \{ \sum_t ((-1)^t/(2t-2j-2k-3)) \binom{b}{t} (w^{j+k}/(n+2j+2k) - w^{t-3/2}/(n+2t-3)) \\ - \sum_{t, t \neq \frac{1}{2}(\ell+1)} ((-1)^t/(2t-\ell-1)) \binom{b}{t} (w^{\frac{1}{2}\ell-1}/(n+\ell-2) - w^{t-3/2}/(n-2t-3)) \}$$

where

$$(2.27) \quad K = (B(n-2, m) B(\frac{1}{2}n, \frac{1}{2}m) B(\frac{1}{2}(n-3), \frac{1}{2}m))^{-1}$$

and

$$(2.28) \quad f(\ell, t, k) = \binom{m-1}{\ell} \binom{b}{t} \binom{b}{k}$$

Case 4

For $p = 5$ the density of W is given by

$$(2.29) \quad Ke^{-\frac{1}{2}\lambda^2} w^{\frac{1}{2}n-1} \sum_{j=0}^{\infty} a_j (T_{1j} + 4T_{2j} + 8T_{3j})$$

where

$$(2.30) \quad T_{1j} = \sum_{k=0}^b (-1)^k f(2k+2j+2, o, k) w^{j+k} \{ ({}_{2j+2k+4}^{m-1} \log w)^2 \\ + 4 \sum_{t \neq 2j+2k+4} (-1)^t / (t-2j-2k-4) ({}_t^{m-1}) \\ \cdot (2w^{\frac{1}{2}t-j-k-2} - 2) / (t-2j-2k-4) - \log w \}$$

and

$$(2.31) \quad T_{2j} = \sum_k \sum_{\ell \neq 2j+2k+2} ((-1)^k / (\ell-2j-2k-2)) f(\ell, o, k) \\ \cdot w^{j+k} \log w \{ ({}_{\ell+2}^{m-1} w^{\frac{1}{2}\ell-j-k-1} - (-1)^\ell ({}_{2j+2k+4}^{m-1}) \}$$

and

$$(2.32) \quad T_{3j} = \sum_k \sum_{\ell \neq 2j+2k+4} ((-1)^{\ell+k} / (\ell-2j-2k-2)) f(\ell, o, k) \\ \cdot \{ \sum_{t \neq 2j+2k+4} (-1)^t ({}_t^{m-1}) w^{j+k} - w^{\frac{1}{2}t-2} / (t-2j-2k-4) \\ - \sum_{t \neq \ell+2} (-1)^t ({}_t^{m-1}) w^{\frac{1}{2}\ell-1} - w^{\frac{1}{2}t-2} / (t-\ell-2) \}$$

and the cumulative distribution function is given by

$$(2.33) \quad 2Ke^{-\frac{1}{2}\lambda^2} w^{\frac{1}{2}n} \sum_{j=0}^{\infty} (T_{1j}^* + 4T_{2j}^* + 8T_{3j}^*)$$

where

$$(2.34) \quad T_{1j}^* = \sum_{k=0}^b (-1)^k f(2j+2k+2, o, k) w^{j+k} [(1/(n+2j+2k)^3) \\ ({}_{2j+2k+4}^{m-1}) \{ ((n+2j+2k) \log w - 2)^2 + 4 \} \\ + 4 \sum_{t \neq 2j+2k+4} ((-1)^t / (t-2j-2k-4)) ({}_t^{m-1}) (1/(n+2j+2k)^2) \\ \{ \{ 2((n+2j+2k)^2 / (n+t-4)) w^{\frac{1}{2}t-j-k-4} - 2(n+2j+2k) \} / (t-2j-2k-4) \\ - (n+2j+2k) \log w + 2 \}]$$

and

$$\begin{aligned}
 (2.35) \quad T_{2j}^* &= \sum_{\ell, k, \ell \neq 2j+2k+2} ((-1)^{k/(\ell-2j-2k-2)}) f(\ell, o, k) w^{j+k} \\
 &\cdot \{ (1 / \{ (n+\ell-2)(n+2j+2k) \}) \log w \cdot \{ (n+2j+2k) \binom{m-1}{\ell+2} w^{\frac{1}{2}\ell-j-k-1} - (-1)^\ell \\
 &\cdot \binom{m-1}{2j+2k+4} (n+\ell-2) \} - (2 / \{ (n+\ell-2)^2 (n+2j+2k)^2 \}) \\
 &\cdot \{ (n+2j+2k)^2 w^{\frac{1}{2}\ell-j-k-1} \binom{m-1}{\ell+2} - (-1)^\ell \binom{m-1}{2j+2k+4} (n+\ell-2)^2 \} \}
 \end{aligned}$$

where

$$\begin{aligned}
 (2.36) \quad T_{3j}^* &= \sum_k \sum_{\ell, \ell \neq 2j+2k+2} ((-1)^{\ell+k/(\ell-2j-2k-2)}) f(\ell, o, k) \\
 &\cdot \{ \sum_{t, t \neq 2j+2k+4} (-1)^t \binom{m-1}{t} w^{j+k/(n+2j+2k)} - w^{\frac{1}{2}t-2/(n+t-4)} / \\
 &\cdot (t-2j-2k-4) - \sum_{t, t \neq \ell+2} (-1)^t \binom{m-1}{t} w^{\frac{1}{2}\ell-1/(n+\ell-2)} \\
 &- w^{\frac{1}{2}t-2/(n+t-4)} T(t-\ell-2) \} , \quad 0 \leq w \leq 1
 \end{aligned}$$

where

$$(2.37) \quad K = (8B(\frac{1}{2}n, \frac{1}{2}m)B(n-2, m)B(n-4, m))^{-1}$$

3. Noncentral distributions of $|R|$ and $|I-R|$

The generalised multiple correlation matrix R plays an important role in testing independence of sets of variables (Khatri (1964), Troskie (1966)). Several test criteria are based on R , i.e. $\text{tr } R$, $\text{tr } R(I-R)^{-1}$, $|R|$ and the likelihood ratio criterion $|I-R|$. In this section we will derive the densities of $|R|$ and $|I-R|$ for the linear case, that is when there is only one nonzero population canonical correlation coefficient.

The noncentral moments of $|R|$ for the linear case are given by Troskie (1969)

$$\begin{aligned}
 (3.1) \quad E|R|^h &= (\Gamma_q(\frac{1}{2}n)\Gamma_q(\frac{1}{2}r+h)/(\Gamma_q(\frac{1}{2}n+h)\Gamma_q(\frac{1}{2}r)))(1-\rho_1^2)^{\frac{1}{2}n} \\
 &\quad {}_3F_2(\frac{1}{2}n, \frac{1}{2}n, \frac{1}{2}r+h; \frac{1}{2}n+h, \frac{1}{2}r; \rho_1^2) \\
 &= E(\prod_{i=1}^q X_i)^h
 \end{aligned}$$

where X_i ($i=1, 2, \dots, q$) are independent Beta variables with density functions (Troskie (1969))

$$(3.2) \quad (B(\frac{1}{2}n - \frac{1}{2}r; \frac{1}{2}(r-i+1)))^{-1} x_1^{\frac{1}{2}(r-i+1)-1} (1-x_1)^{\frac{1}{2}(n-r)-1}, \quad i=2,3,\dots,q$$

and

$$(3.3) \quad (B(\frac{1}{2}r; \frac{1}{2}(n-r)))^{-1} x_1^{\frac{1}{2}r-1} (1-x_1)^{\frac{1}{2}(n-r)-1} (1-\rho_1^2)^{\frac{1}{2}n} {}_2F_1(\frac{1}{2}n, \frac{1}{2}n; \frac{1}{2}r; \rho_1^2 x_1).$$

Hence the square of the vector correlation coefficient $|R|$, is distributed as the product of q independent Beta variables with density functions (3.2) and (3.3). Denoting $|R|$ by $W_{q,n-r,r}$ it can easily be seen that explicit expressions of the density for the cases $q=2,3,4$ and 5 differ from those of $|I-L|$ only in that

- (i) W now denotes $|R|$
- (ii) $(1-\rho_1^2)^{\frac{1}{2}n}$ replaces $e^{-\frac{1}{2}\lambda^2}$
- (iii) $a_j = (\frac{1}{2}n)_j (\frac{1}{2}n)_j \rho_1^{2j} / ((\frac{1}{2}n)_j j!)$
- (iv) n and m in the expressions for $|I-L|$ are replaced by r and $(n-r)$ respectively.
- (v) $b = \frac{1}{2}(n-r-2)$

The noncentral moments of the likelihood ratio criterion $|I-R|$ for the linear case are given by (Troskie (1969))

$$(3.4) \quad E(|I-R|^h) = (\Gamma_q(\frac{1}{2}n) \Gamma_q(\frac{1}{2}(n-r)+h) / (\Gamma_q(\frac{1}{2}n+h) \Gamma_q(\frac{1}{2}(n-r)))) \\ \cdot (1-\rho_1^2)^{\frac{1}{2}n} {}_2F_1(\frac{1}{2}n, \frac{1}{2}n; \frac{1}{2}n+h; \rho_1^2) \\ = E(\prod_{i=1}^q X_i)^h$$

where X_i ($i=1,2,\dots,q$) are independent Beta variables with density functions (Troskie (1969))

$$(3.5) \quad (B(\frac{1}{2}(n-r), \frac{1}{2}(n-r-i+1)))^{-1} x_1^{\frac{1}{2}(n-r-i+1)-1} (1-x_1)^{\frac{1}{2}r-1} \quad i = 2,3,\dots,q$$

and

$$(3.6) \quad (B(\frac{1}{2}(n-r); \frac{1}{2}r))^{-1} x_1^{\frac{1}{2}(n-r)-1} (1-x_1)^{\frac{1}{2}r-1} (1-\rho_1^2)^{\frac{1}{2}n} {}_2F_1(\frac{1}{2}n, \frac{1}{2}n; \frac{1}{2}r; \rho_1^2 (1-x_1))$$

The explicit expressions for $|I-R|$ will coincide with those of $|L|$ derived by Gupta except for the following changes

- (i) U now denotes $|I-R|$
- (ii) $(1-\rho_1^2)^{\frac{1}{2}n}$ replaces $e^{-\frac{1}{2}\lambda^2}$
- (iii) f_1 and f_2 are replaced by $(n-r)$ and r respectively
- (iv) $a_j = (\frac{1}{2}n)_j (\frac{1}{2}n)_j \rho_1^{2j} / ((\frac{1}{2}n)_{jj}!)$
- (v) $b = \frac{1}{2}(r-2)$.

4. Complex Analogues of $|L|$, $|I-L|$, $|I-\sum_j L_j|$, $|R|$ and $|I-R|$

De Waal (1968) and Troskie (1969) have shown that the distributions (noncentral linear case) of the complex analogues of the real variable statistics, $|L|$, $|I-L|$, $|I-\sum_j L_j|$, $|R|$ and $|I-R|$, are the same as the product of p (say) beta variables, one being distributed as a noncentral beta and the remaining $(p-1)$ being distributed as central beta's, the value of the parameters being twice those for the corresponding real variable case. Hence it is impossible to use the duplication formula on the moments of these statistics and as a consequence theorem 2.2 is no longer applicable.

Explicit expressions for the density functions and cumulative distribution functions of these statistics can still be obtained by the method of convolutions. However since it is not possible to invoke a theorem similar to theorem 2.2, the expressions already become very complicated for $p > 3$. Expression for $p = 2$ and $p = 3$ are given in Money (1972).

On the credit side it will be noticed that in the complex case the derivations of the expressions involve binomial expansions of the form $(1-x)^a$ where a is integral, and hence the tabulation of the cumulative distribution function for all m and n is possible by means of the same algorithm.

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